Modelling Free-Fall with Varying Drag

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INTRODUCTION

On 14th of October 2012, Felix Baumgartner set the world record for the furthest free-fall ever done by a human, he jumped from the Earth's stratosphere, $38,969.4m$ up above where the temperature is as low as $-70.9°C$.His record breaking skydive took 260 seconds until he deployed his parachute to land.[1] But his little adventure was watched by billions around the world. The main objective of this report is to analyse by creating a model of a free-falling object due to gravitational acceleration and simulating its motion using Newtonian mechanics, and to apply this knowledge of Newtonian mechanics with Euler's method for free-fall under gravity, in order to simulate Baumgartner's fall. The Methods will be divided into Parts A, B and C, which get more complex as we progress.

THEORY

When travelling through air, any free-falling object with velocity v, will experience air resistance, drag force F_{drag} in the opposite direction to its movement,

$$F_{drag} = -k|v|v \tag{1}$$

Where k is the drag constant and is given by:

$$k = \frac{C_d \rho_0 A}{2} \tag{2}$$

Where C_d is the drag coefficient, ρ_0 is the air density and A is the cross-sectional area of the object. At atmospheric pressure and temperature of $273K$, $\rho_0 = 1.293kgm^{-3}$. Considering the air density change as a function of altitude y, $\rho(y)$ can be described as:

$$\rho(y) = \rho_0 e^{-y/h} \tag{3}$$

Where $-y/h$ is the ratio of altitude to the atmosphere's scale height, *h* which is equal to $7.64km^2$. Substituting *eq.3* to *eq.2*, we get:

$$k(y) = \frac{C_d \rho_0 e^{-y/h} A}{2} \tag{4}$$

Excluding the horizontal forces, the vertical forces acting on the sky diver are gravity hence their weight, W acting down, and F_{drag} acting upwards. From Newton's Second Law, $F = ma = m\frac{dv_y}{dt} = W + F_{drag}$, we get:

$$m\frac{dv_y}{dt} = -mg - k|v_y|v_y \tag{5}$$

Where g is the acceleration caused by gravity and is $9.81ms^{-2}$. Taking y-coordinate as positive upwards and negative downwards, which makes v_y is a negative vector since the motion is downwards. Where,

$$\frac{dy}{dt} = v_y = f(y,t) \tag{6}$$

Euler's method for solving the free-fall problem can be summarised as:

$$v_{y,n+1} = v_{y,n} - \Delta t(g + \frac{k}{m}|v_{y,n}|v_{y,n}) \tag{7}$$

$$t_{n+1} = t_n + \Delta t \tag{8}$$

$$y_{n+1} = y_n + v_{y,n}\Delta t \tag{9}$$

Where we determine v, t and y at the $(n+1)^{th}$ step from their values at the previous (n^{th}) step.

METHOD

For Part A, by solving the *equations 5* and *6* we get:

$$y = y_0 - \frac{m}{k} log_e \left[cosh\left(\sqrt{\frac{kg}{m}}t\right)\right] \tag{10}$$

$$v_y = -\sqrt{\frac{mg}{k}} tanh\left(\sqrt{\frac{kg}{m}}t\right) \tag{11}$$

Part A was quite straight forward, there was 200 equally spaced values for t for the first 10 seconds of the fall (hence the time step, dt=0.05s) in order to calculate y and v_y using *eqs.10* and *11*.
y_0 is the initial drop height of the object and for Part A this was set to 1000m. When calculating the drag constant, k, the change in air density was not considered, therefore k was calculated with *eq.2*.

Parts B and C were calculated using Euler's method with *eq.7-9*, which improved on Part A. In Part B, k was calculated using *eq.2*, which introduced the effects of C_d, ρ_0, and

A, and Part C improved on Part B by using *eq.4*, introducing $e^{-y/h}$ into the equation, hence the effect of the change in altitude. Making ρ, therefor k a function of altitude is expected to improve the accuracy of the result since now F_{drag} is also a function of altitude.

When comparing Part A to other methods and in Parts B and C, the total mass of the skydiver is $m_{total} = m_{diver} + m_{suit}$, where Felix Baumgartner's mass taken as $m_{diver} = 73kg$, and $m_{suit} = 31.7515kg$.

Assuming the cross-sectional area A of a skydiver facing down to have a rectengular shape. When estimating Baumgartner's cross-sectional area wearing the suit, 0.05m were added both to his height (1.73m) and width (assuming average male shoulder to shoulder width to be 0.5m) which gave an estimate of $A = (1.73 + 0.05)(0.5 + 0.05) = 0.9515m^2$.[2, 3]

Drag coefficient, C_d of a skydiver was taken as 1.0 for all calculations that were used to compute values for y and v_y, but realistically this value would vary since it is dependent on other parameters, such as; the speed of the diver relative to the fluid (air), changes in air pressure, etc. The speed of the diver is especially important because the C_d of 1.0 is for a regular skydiver, whereas in Baumgartner's case, because he reaches speeds greater than the speed of sound that regular skydivers don't reach. Therefore, C_d for Baumgartner would vary over the duration of the fall and be substantially different.

RESULTS

Part A:

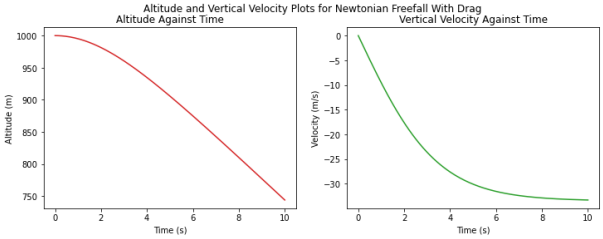


FIG. 1. Altitude and vertical velocity against time plots for the first 10 seconds of a free-fall, where $m = 80kg$, $k = 0.7kgm^{-1}$, and $y_0 = 1000m$.

Figure 1 shows that as the $|v|$ of the falling object increases, F_{drag} acting on the body also increases causing the object to reach a terminal velocity.

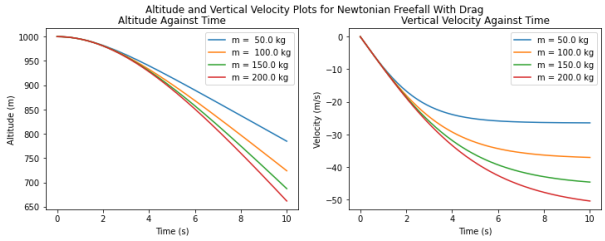


FIG. 2. Altitude and vertical velocity against time plots for the first 10 seconds of a free-fall, that compares various masses, where $k = 0.7kgm^{-1}$, and $y_0 = 1000m$.

Fig. 2 shows us the effect of mass on free-fall. The object with greater mass will reach a higher v_{max}. This is because the downward force acting on the body *W=mg*will be bigger, which means a greater F_{drag} is needed for the object to reach its terminal velocity, hence the greater v_{max}.

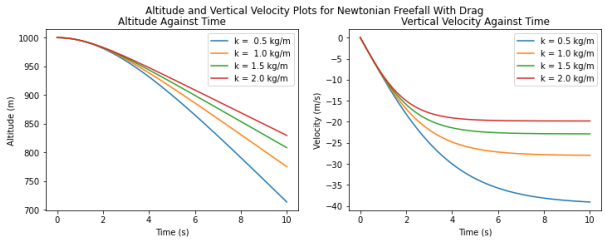


FIG. 3. Altitude and vertical velocity against time plots for the first 10 seconds of a free-fall, that compares various constant drag factors, where $m = 80kg$, and $y_0 = 1000m$.

Fig. 3 shows us the effect of the drag constant, k on free-fall. A greater k will result in a higher v_{max}. Since the masses, therefore $W = mg$ in this case are all the same. If k is greater, v_{max} has to be lower in order to balance the equation.

Part B:

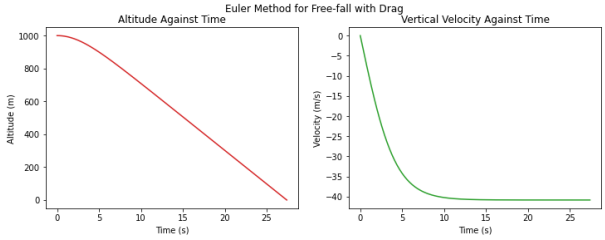


FIG. 4. Altitude and vertical velocity against time plots of a free-fall using Euler's method, with $m = 104.7515kg$, $C_d = 1$, $A = 0.9515m^2$, and $y_0 = 1000m$.

In *fig. 4*, when all the parameters were set, an output value of $k = 0.61514475$ was obtained. The object's took 27.4 seconds to reach the ground, and its maximum velocity was $v_{max} = -40.87m/s$.

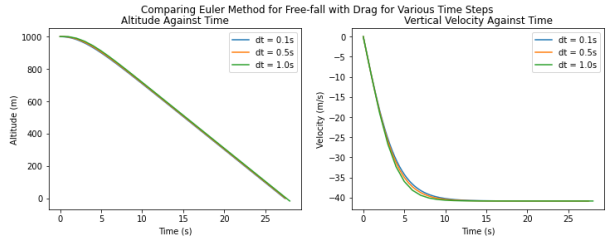


FIG. 5. Altitude and vertical velocity against time plots of a free-fall using Euler's method, that compares the use of different time steps, with $m = 104.7515kg$, $C_d = 1$, $A = 0.9515m^2$, $k = 0.61514475$ and $y_0 = 1000m$.

In *fig. 5*, the effect of the time step, dt which is used to calculate values for y and v_y can be observed. Greater values of dt causes the plot to be more rough. Looking at *fig. 5*, it is clearly seen that the smoothest plot has a time step of $dt = 0.1s$. A bigger dt means that the simulation will be updated less regularly which will lead to inaccuracies in our calculations.

Comparing Parts A & B:

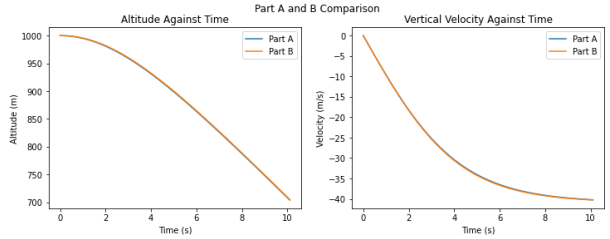


FIG. 6. Altitude and vertical velocity against time plots of the first 10 seconds of free-fall that compares Parts A and B, with $m = 104.7515kg$, $C_d = 1$, $A = 0.9515m^2$, $k = 0.61514475$, $dt = 0.5s$ and $y_0 = 1000m$.

For Parts A and B, the altitudes reached at $t = 10s$ is *687.779m* and *692.421m* respectively, the difference between the altitudes reached is *4.642m*. The v_{max} reached were $v_{max,A} = 40.24m/s$ and $v_{max,B} = 40.29m/s$, therefore the difference in maximum velocity reached, v_{max} between Parts A and B is *0.052m/s*.

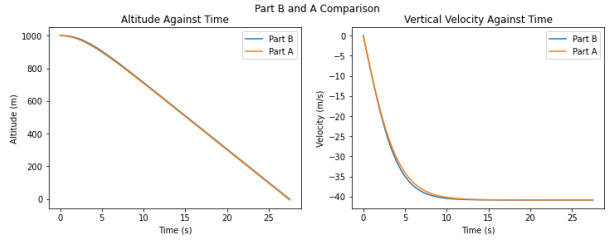


FIG. 7. Altitude and vertical velocity against time plots of a free-fall (until the objects reach $y = 0.0m$) that compares Parts B and A, with $m = 104.7515kg$, $C_d = 1$, $A = 0.9515m^2$, $k = 0.61514475$, $dt = 0.5s$, and $y_0 = 1000m$.

For Parts B and A, the altitudes reached at $t = 20s$ are $304.791m$ and $299.043m$ respectively, with a difference of $5.748m$. The v_{max} reached were $v_{max,A} = 40.24m/s$, the difference in maximum velocity reached, v_{max} between Parts B and A is $1.23x10^{-4}ms$.

Looking at *figures 6* and *7*, it is observed that, when the functions are run for a longer period, the difference in v_{max} decreases but the difference in altitude increases.

Part C:

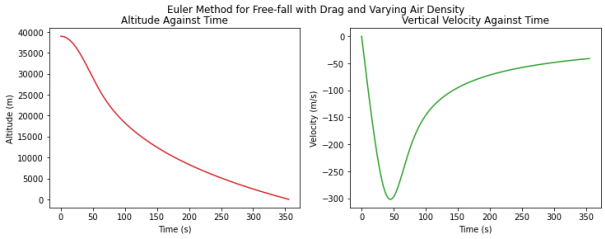


FIG. 8. Altitude and vertical velocity against time plots of Baumgartner's fall, where air density varies as a function of altitude, $\rho(y)$, $m = 104.7515kg$, $C_d = 1$, $A = 0.9515m^2$, and $y_0 = 38969.4m$.

Maximum velocity reached, $v_{max} = -301.86m/s$ and time of flight is $355.4s$. The ratio output is $C_dA/m = 0.00908$.

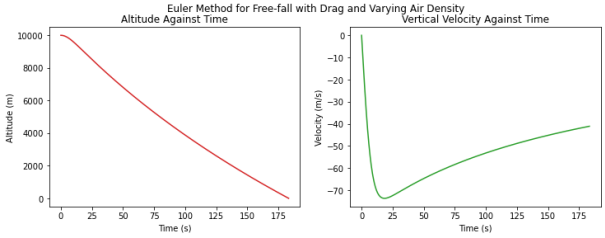


FIG. 9. Altitude and vertical velocity against time plots of Baumgartner's fall, where air density varies as a function of altitude, $\rho(y)$, $m = 104.7515kg$, $C_d = 1$, $A = 0.9515m^2$, and $y_0 = 10000m$.

When all parameters, apart from the initial drop height was kept the same as in *fig.8*, which lets us see the effect of y_0 on v_{max} and the time of flight. The calculations in *fig.9* gave us a maximum velocity of $v_{max} = 73.69m/s$ and a time of flight of $183.2s$. At lower altitudes the air density is not as low as it is at greater altitudes, which means k is higher (more air particles to resist the falling body), hence the lower value for v_{max}.

Part C, Change in C_dA/m:

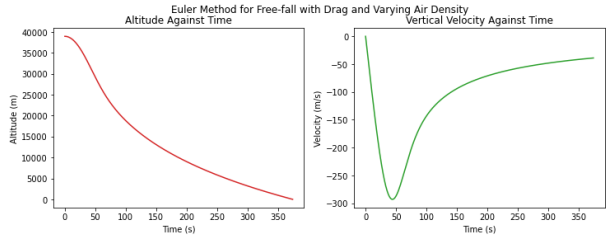


FIG. 10. Altitude and vertical velocity against time plots of free-fall, where air density varies as a function of altitude, $\rho(y)$. $C_dA/m = 0.01011$, $y_0 = 38969.4m$.

In *fig.10* maximum velocity reached is, $v_{max} = -293.04m/s$ and time of flight is $373.9s$.

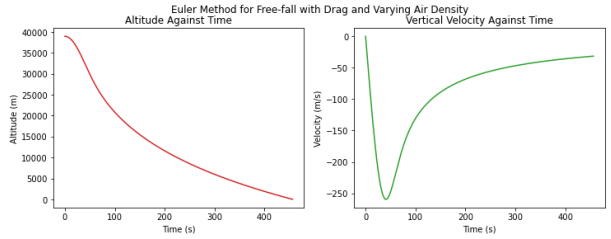


FIG. 11. Altitude and vertical velocity against time plots of free-fall, where air density varies as a function of altitude, $\rho(y)$. $C_dA/m = 0.0153$, $y_0 = 38969.4m$.

In *fig.11* maximum velocity reached is, $v_{max} = -259.64m/s$ and time of flight is $456.4s$.

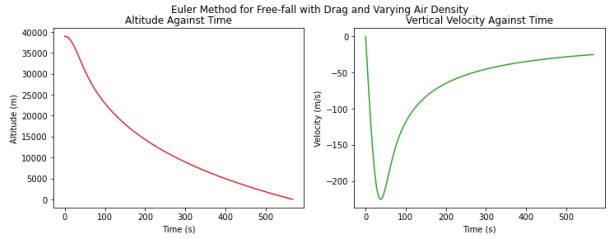


FIG. 12. Altitude and vertical velocity against time plots of free-fall, where air density varies as a function of altitude, $\rho(y)$. $C_dA/m = 0.024$, $y_0 = 38969.4m$.

In *fig.12* maximum velocity reached is, $v_{max} = -225.56m/s$ and time of flight is $567.3s$.

C_dA/m (x10^{-2})	v_{max}(m/s)	time of flight, t (s)
0.908	-301.86	355.4
1.011	-293.04	373.9
1.530	-259.64	456.4
2.400	-225.56	567.3

TABLE I. A table of C_dA/m, time of flight (s), and v_{max}(m/s).

Table I shows clearly that as the C_dA/m ratio increases, v_{max} decreases accordingly, which leads to a longer time of flight, t.

DISCUSSION AND ANALYSIS

The analytical models made to calculate altitude, y and vertical velocity, v_y in Parts A, B and C all proved effective when simulating the free-fall of an object. When methods for A and B, are compared it is safe to say that the Euler's method (B) proved more effective since we were able to set a desired dt which allowed us to make calculations that were updated more regularly, therefore more accurate.

Figures 6 and *7* gave us the differences in altitude and v_{max}. These differences in altitude and v_{max} were: $dy_{AB} = 4.642m$ and $dv_{max,AB} = 0.052m/s$ at $t = 10s$, and $dy_{BA} = 5.748m$ and $dv_{max,BA} = 1.23x10^{-4}m/s$ at $t = 20s$. Percentage difference of these 4 different values are; $dy_{AB,100} = 0.00670\%$, $dv_{max,AB,100} = 0.5197\%$, and $dy_{BA,100} = 1.8861\%$, $dv_{max,BA,100} = 0.0003\%$. These percentage differences tell us that the differences in altitude get bigger as the function runs longer and the percentage error on it also gets bigger with it, whereas unlike the altitude v_{max} and its percentage difference gets smaller as the function runs for longer.

For Part C, *fig.8* showed a rapid increase in velocity, due to the high initial altitudes since the air is less dense at such heights than lower heights. A maximum velocity of $v_{max} = 301.86m/s$, and a time of flight of $355.4s$ is observed. Both of these values are quite a lot greater than the velocities that were reached in Parts A and B. In Part C, when the effects of certain parameters were observed. Keeping everything apart from the initial height the same, proved that as the initial altitude decreases, the v_{max} reached decreases. When other parameters that would vary the ratio C_dA/m are changed, it was seen that as C_dA/m increased, v_{max}, and therefore the time of flight decreased accordingly.

Errors and Improvements

Apart from the improvements in the code which can also be improved, the models could include the simulation of a 3D human model, a more capable simulation software used in engineering and fluid-dynamics. None of the methods discussed in this report took the diver's parachute into consideration, since Braumgartner opened his parachute thousands of metres above the ground, he experienced a massive decrease in his velocity, which affected his time of flight. The height of the atmosphere h varies anywhere between 7.640km to 8.5km, if different values of h and the improvements mentioned above are taken into consideration in the future, a more accurate and precise simulation of free-fall can definitely be achieved.

REFERENCES

[1] RedBull Stratos [online] Available at: www.redbull.com/gb-en/projects/red-bull-stratos [Accessed: November 2023]

[2] Felix Baumgartner, Bio, [online] Available at: felixbaumgartner.com/bio [Accessed: November 2023]

[3] G.P. Thomas, Skydive from Space - Felix Baumgartner's Suit, [online] Available at: www.azom.com/article.aspx?ArticleID=7576#:~:text=The%20total%20weight%20of%20the,3.5%20pounds%20per%20inch2) [Accessed: November 2023]